

CURBRISK

Find the puddle before it floods the basement.

Curb-level drainage risk, surveyed in days — not a FEMA cycle.

Existing flood maps miss where streets actually fail.

FOR INSURERS

FEMA maps are coarse and stale.

FEMA flood maps refresh on 5–10 year cycles and average risk across entire zones. The crest house and the low-point house on the same block carry the same premium — and most flood losses happen outside the official zone, at street drainage failures the map never sees.

FOR CITIES

Street drainage gaps stay invisible.

Undersized basins, missing inlets, and ponding pockets don't appear on any flood map. Cities only learn about them through 311 complaints, after basements flood.

Streets get repaved over unresolved drainage and fail again in 18 months — and 1970s-era basins now carry 2020s-era storm loads.

Find every street drainage gap — in days, not years.

LiDAR

53.5M points
road-surface DEM

Hydrology

low-point detection,
catchment + 2-in ponding

Score

topo 40 / PCI 30 /
basin 20 / 311 10

Deliver

APS cross-section
+ address API

WHAT COMES OUT

Every drainage gap on every street, scored 0–100 per 30-ft segment, with a browser-viewable 3D cross-section of each low point. A city in weeks, not a FEMA cycle.

Demo footprint: 1 Somerville tile • 5,080 pavement sections • 381 catch basins • 145 flooding 311s

Faster, finer, and built around the street itself.

Days, not decades

FEMA flood maps refresh on 5–10 year cycles. Cyvl mobile LiDAR scans a city in a single drive; CurbRisk re-scores the network in days. Drainage projects don't have to wait on a federal map update.

Curb-level, not zone-level

FEMA averages risk across whole zones. CurbRisk scores every 30-ft pavement segment and every detected low point — measured terrain, local impervious cover, real complaint history.

Street drainage gaps, not just floodplains

Identifies undersized basins, missing inlets, ponding pockets, and flow-concentration points — the failure mode that floods basements outside the official flood zone, that no flood map shows.

Validated + Autodesk-delivered

144 Somerville flooding 311s ground-truth the model. Every flagged location exports as a browser-viewable APS cross-section — engineers in Civil 3D, underwriters in a browser.

Where FEMA gives you a polygon every 5 years, CurbRisk gives you every drainage gap on every street, this month.

Two markets, one engine.

PRIMARY • INSURANCE & RISK

Address-level flood pricing

- Underwriters: per-query API at point of sale
- Actuaries: portfolio loss exposure below the FEMA zone
- Cat modelers (RMS, Verisk): bulk dataset license

Differentiates premiums between the crest house and the low-point house on the same block.

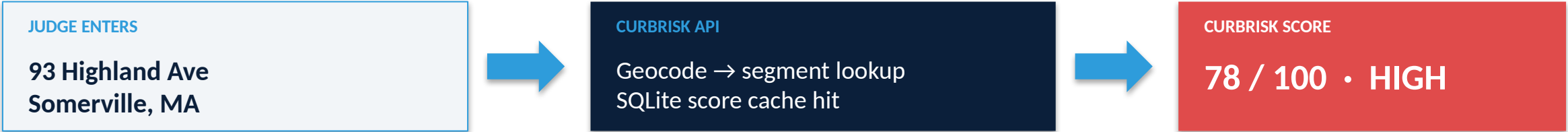
SECONDARY • MUNICIPAL DPW

Defensible capital prioritization

- Stop repaving over unresolved drainage
- Auto-generate MS4 / NPDES gap analysis
- Hand civil engineers a Civil 3D starting model
- Move to the front of FEMA BRIC / EPA CWSRF queues

Replaces complaint-driven politics with a ranked, auditable list.

Address in. Drainage risk out. Under 10 seconds.



RESPONSE PAYLOAD	
• Score breakdown	topo 32 • pavement 21 • basin 18 • complaints 7
• LiDAR geometry	low-point depth, 2-in ponding volume, catchment polygon
• 311 history	3 flooding reports within 100 m since 2023
• APS cross-section	browser-viewable 3D profile — no CAD software needed
• Recommended action	clear upstream basin; do not resurface until drainage resolved

WHAT'S NEXT

From one Somerville tile to a city-wide risk layer.

NOW

Hackathon MVP

One Somerville tile end-to-end. Address → score + APS cross-section in < 10 s.

NEXT

Paid pilot

\$20–50K fixed-scope study with a Somerville-style city. One paving package, one Autodesk model, one reference customer.

THEN

Insurer dataset

Bulk drainage-risk layer for one metro area, licensed to a cat modeler. Recurring revenue, no per-city sales cycle.

Stop waiting on a federal map cycle to find a failing catch basin.

Score every street drainage gap, every block, every month.